If Two only

$2'$ { V C, V R, V Δ, V L } +12 bits

V_{25} 25

9. in Exchange

IBM can decide between #3 & #2' or a scheme which includes these Two.

(Brooks)

Scheme:

No immediate index modif. w > 19 bits → dir. index mod gets 1 more bit

All index mod. instr. are indexable & take ½ wd.

Four bit index addr.

all index modif. values are full wd → direct index mod. gets a 2nd bit.

all present direct index mod. ops.

Now index mod. ops.

15 8	13	1	load $\frac{L}{R}$ II	from R(A)/L(A)	
16 17	14	2	store $\frac{R}{L}$ II	at R(A)/L(A)	
	19	3	Comp $\frac{L}{R}$ II	w L(A)/R(A)	
	20	4	Add		
	$2'_5$		Sub		
	18		Incr + Br. to C(A)	Δ	

9. Incr. # by C(A) + Ct/not		C
10. Incr	C(A) + Compare	L
11. Comp II	+ Reset	R
12 Ct. + Br.	C(A) + Reset	R

(32 max can be done)

in MISC. format

Load IL immed (A)
IR (A)
Add IL " A
IR A
Sub IL " A
R A
Comp IL " A
R A

all opt with sign mod. 6.91%

other { 8.4% static
7.3 dyn.
2.5 time

Comparison of Schemes.

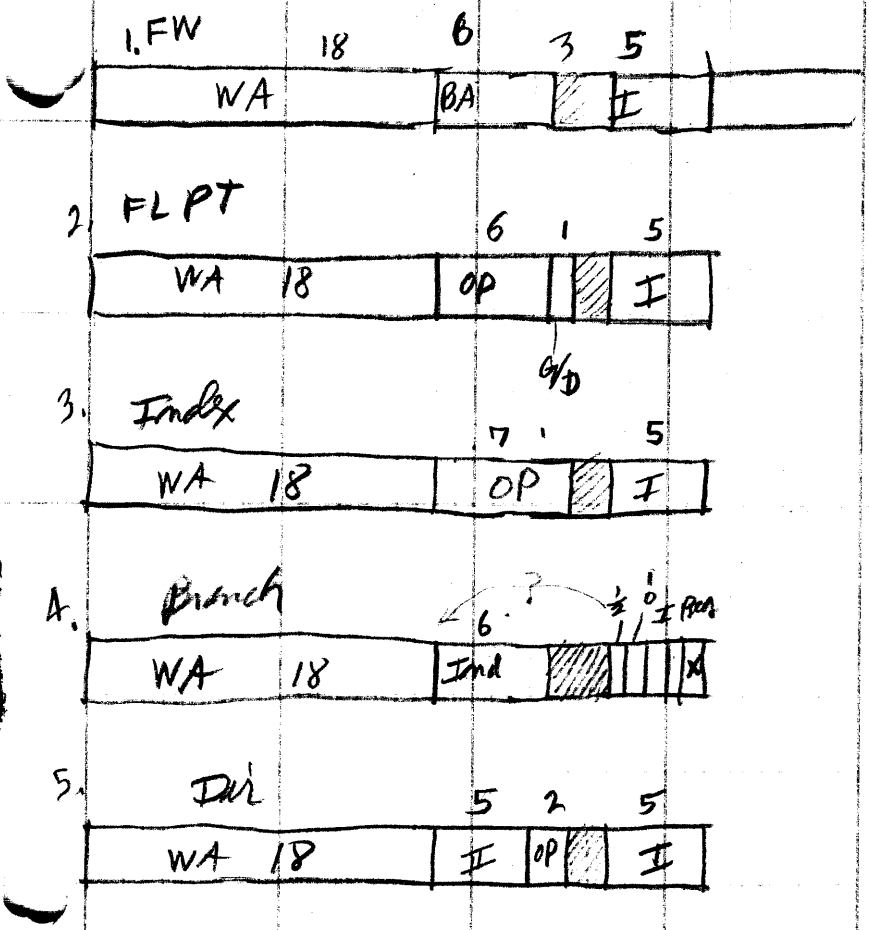
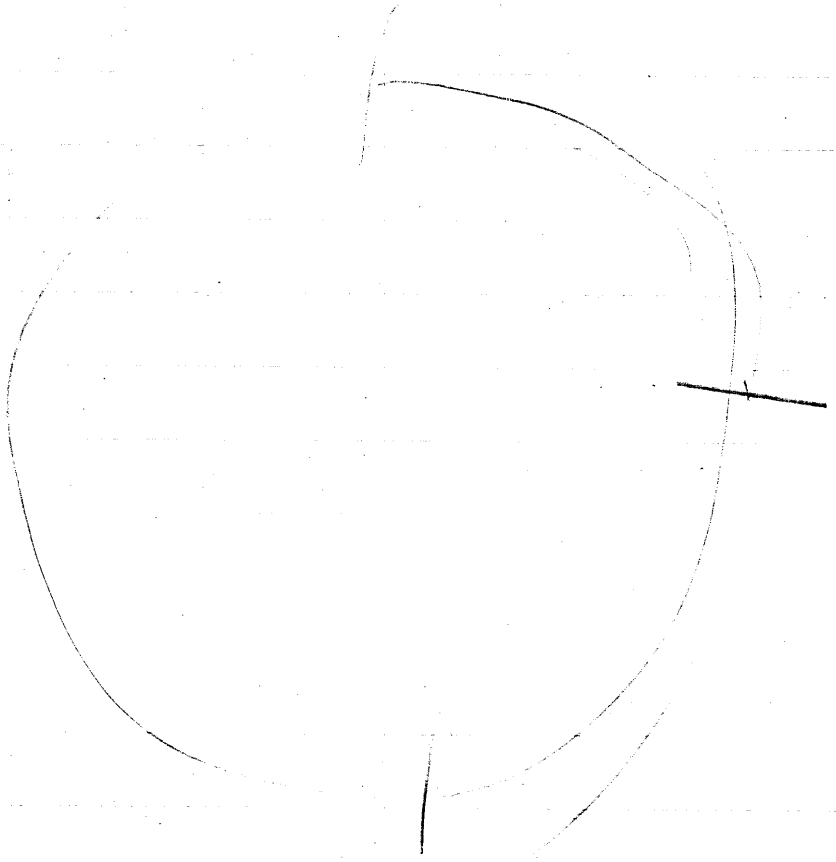
LA HW Compromise (5 bit)	LA FW	IBM HW (full word index)
(31) 5 Register bits	6 bit.	Bit Eff gain 220%
Limited 2nd ind	Full Index Indexing	Good Indexed indexing
High Bit Eff. (20%)	Imm. bit Indexing	15 registers
Simplicity	Large vocab.	Simplicity
64 floating op. (128) without geo.	Full capacity for storing transfer addr.	192 Floating
Index "shift"	Indirect addressing.	Index "shift"
	Dir. geometric	Limited transfer store
	2048 Fl Pt	
1 Branch	1 Branch	1 Branch
4 direct		8 semi index
		32

Full Wd. Larger no. of index regs.

assume

Preload
Post store
Store + Load.

Investigate full wd index (2 type) IB decided on 3,
1. 4 bit V, L, R
2 Full Wd.

Decision by Thursday

Los Alamos "2am" 5 bit code